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ABC-XYZ Classification and Forecasting for Inventory Optimization

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ABSTRACT

Effective inventory management remains a significant challenge for businesses facing fluctuating demand and seasonal variations. Traditional methods often fall short, leading to issues such as stockouts, overstocking, and inefficient resource use. This study presents an integrated approach to inventory management by combining ABC-XYZ classification with time series forecasting using machine learning techniques. The ABC-XYZ analysis categorizes inventory items based on their revenue significance and demand variability, enabling a more nuanced understanding of inventory prioritization. By leveraging historical sales data, we apply Random Forest Regression to forecast future demand for different item categories, providing actionable insights for optimizing inventory levels. This approach not only enhances the accuracy of inventory predictions but also supports more efficient resource allocation, ultimately reducing stockouts and overstock situations. The findings demonstrate the potential of advanced analytics in streamlining inventory management processes and improving overall supply chain efficiency.

Keywords: Inventory Management, ABC-XYZ Classification, Machine Learning, Random Forest Regression, Demand Variability, Sales Prediction

INTRODUCTION

Effective inventory management is a cornerstone of operational efficiency and profitability across various industries. As businesses strive to meet customer demands while minimizing costs, the role of inventory management becomes increasingly critical. At its core, inventory management involves the strategic oversight of stock levels, ensuring that goods are available to meet customer needs without incurring unnecessary holding costs. In today's rapidly evolving market landscape, characterized by fluctuating demand, supply chain complexities, and the proliferation of digital technologies, the need for sophisticated inventory management techniques has never been more pressing.

The evolution of inventory management reflects broader trends in business and technology. Traditional inventory management practices, which often relied on manual tracking and rule-of-thumb estimations, are being supplanted by data-driven approaches that leverage advanced analytics and machine learning (ML). Among these modern techniques, ABC-XYZ classification has emerged as a powerful tool for categorizing inventory based on both value and variability, enabling businesses to prioritize their management efforts more effectively.

ABC-XYZ classification offers a nuanced approach to inventory management by segmenting inventory items into categories based on their contribution to overall revenue (ABC) and the consistency of their demand (XYZ). This dual classification system allows businesses to tailor their inventory strategies to the specific characteristics of each item, ensuring that high-value, stable-demand items receive the attention they deserve, while less critical items are managed with a different focus. The integration of ABC and XYZ classification provides a comprehensive framework for understanding the relative importance of inventory items and their impact on overall business performance.

In addition to categorization, forecasting plays a crucial role in inventory management. Accurate demand forecasting enables businesses to anticipate future inventory needs, reducing the risk of stockouts or overstocking. Traditional forecasting methods, while useful, often fall short in capturing the complex, non-linear patterns present in modern supply chains. This is where machine learning, rather than traditional methods, offers significant advantages in forecasting accuracy. By integrating ABC-XYZ classification with ML-based forecasting, businesses can achieve a higher level of precision in inventory management, aligning stock levels more closely with actual demand.

In the context of the manufacturing sector, efficient inventory management is crucial for balancing supply and demand, minimizing costs, and ensuring timely product availability. Manufacturers face significant challenges, including dealing with fluctuating demand, seasonal variations, and the risk of stockouts or overstocking. The integration of ABC-XYZ classification with ML-based forecasting provides a strategic approach to overcoming these challenges, enabling manufacturers to optimize inventory levels, reduce carrying costs, and enhance overall operational efficiency.

Today's inventory management landscape is shaped by rapid technological advancements and increasing market demands. Businesses now face the challenge of balancing inventory levels with fluctuating consumer preferences and supply chain complexities. The integration of digital tools and data analytics has revolutionized inventory management, enabling companies to achieve greater accuracy in forecasting and real-time monitoring of stock

levels. Modern inventory systems are designed to handle large volumes of data, provide actionable insights, and facilitate dynamic adjustments to inventory strategies.

Moreover, the emphasis on customer satisfaction and efficient resource utilization has driven the adoption of advanced techniques and methodologies. Inventory management today not only focuses on optimizing stock levels but also on enhancing the overall customer experience. Companies are leveraging automation, artificial intelligence, and machine learning to streamline operations and respond swiftly to changes in demand. This evolution reflects a shift towards a more data-driven and customer-centric approach in managing inventory.

Machine learning, a branch of artificial intelligence, has emerged as a powerful tool in predictive analytics, including inventory management. By processing large datasets and identifying intricate relationships, ML models can make accurate predictions about future inventory needs based on historical sales data, seasonal trends, and other factors. The technology facilitates the development of advanced predictive models that can adapt to changing market conditions and improve decision-making. As a result, companies can better manage their inventory levels, reduce costs, and enhance overall operational efficiency.

The integration of machine learning with inventory management systems represents a significant advancement in optimizing stock control and meeting customer demands. In this research, we explore the application of ABC-XYZ classification and ML-based forecasting in inventory management. By combining these approaches, we aim to develop a robust framework for optimizing inventory levels, enhancing forecasting accuracy, and improving overall supply chain performance.

This study contributes to the growing body of literature on inventory management by demonstrating the practical application of advanced analytical techniques in real-world settings. Through a detailed analysis of sales data and demand patterns, we highlight the potential of ABC-XYZ classification and machine learning in driving inventory management excellence. As businesses continue to navigate the complexities of the modern market, the insights gained from this research can serve as a valuable guide for implementing more effective and efficient inventory management strategies.

LITERATURE REVIEW

1. **Kehinde Busola and et.al (2020)** – To investigate inventory management practices at Covenant Bakery and their impact on economic performance, focusing on the importance of inventory items and determining the Economic Order Quantity (EOQ) for key items. Using ABC analysis and EOQ techniques, the study finds that to minimize total costs, Covenant Bakery should purchase specific quantities of flour, sugar, and butter per order. It recommends maintaining tight control over these critical items while allowing more flexibility for other inventory categories.
2. **Omid Abdolazimi and et.al (2021)** – To present an advanced mathematical model to enhance traditional ABC analysis by optimizing inventory grouping and control decisions, focusing on maximizing net profit while considering inventory costs and service levels. Using Benders decomposition and Lagrange relaxation algorithms, the study compares solutions through the TOPSIS technique and sensitivity analyses. The approach is validated through a real-world case study in the ceramic tile industry, offering valuable insights for inventory managers on cost-effective and efficient inventory management strategies.
3. **Michael Siek and Kevin Guswanto (2021)**- Explores the development of predictive models for optimal inventory management in the food and beverage industry, focusing on restaurant ingredient management. By applying machine learning algorithms such as linear regression, multi-layer perceptron, random trees, random forests, and model trees to time series data, the study aims to enhance prediction accuracy. Effective use of these models can significantly reduce inventory costs and ingredient waste, ultimately improving food quality and maximizing profit. The research highlights the potential of computational intelligence in addressing the complexities of inventory management.
4. **S. Reddy Gayam and et.al (2021)** - Explores the transformative impact of artificial intelligence (AI) on supply chain management, focusing on predictive maintenance, demand forecasting, and inventory optimization. AI enhances predictive maintenance by using sensor data to foresee equipment failures and reduce downtime. It improves demand forecasting by analysing extensive datasets to generate accurate, dynamic forecasts. AI also optimizes inventory levels through advanced algorithms, minimizing carrying costs and stockouts. Case studies from aerospace manufacturing and retail illustrate AI's effectiveness in enhancing operational efficiency and customer satisfaction, highlighting its potential to revolutionize supply chain management.³
5. **Walid Emar and et.al (2021)** - To examine the inventory management of computer spare parts (CSPs) at Power-One Jordan Computer Hardware-Software Company, focusing on slow-moving items. Key factors affecting CSP management include production costs, obsolescence, availability, and transportation costs. The study forecasts a 20% increase in demand for adapters and chargers using the EOQ model, highlighting a 48% profit margin that necessitates intervention to avoid losses. The research identifies limitations in traditional ABC classification for slow-moving CSPs, recommending an improved ABC model that considers the criticality of these items for better inventory management.
6. **Kaustubh Vaman Sakhare and Isha Kulkarni (2022)** - Addresses issues in order management for essential perishable goods, which often suffer from inaccurate demand estimation. It highlights the identification of key features for estimating replenishment needs, considering factors like calamities and festivities. The study compares complex models using linear programming and reinforcement learning, finding that simpler machine learning algorithms are more effective for small-scale businesses in predicting order estimates under dynamic conditions.

7. **Gudur Shirisha and et.al (2022)** - Streamlines the control and administration of agricultural products from entry to sale. Machine learning-based predictive analytics can optimize processes, such as replacing handwritten accounts and understanding customer behaviour, by forecasting demand based on historical data and seasonal trends. This predictive model helps prevent stockouts and overstocking, maximizing sales potential and profit. The project aims to enhance inventory management for agricultural products through advanced predictive analytics, reducing manual work and improving efficiency.
8. **Patdono Suwignjo and et.al (2023)** - Explores the challenge of understocking and overstocking faced by a major FMCG company in Indonesia. By employing big data and predictive analytics, specifically using a gradient boosting model, the study predicts inventory status and the amount of understock/overstock. The classification model achieved accuracies of 0.84, 0.76, and 0.74 for different product categories, while the regression model attained R^2 values of 0.89, 0.76, and 0.74, respectively. Results are visualized in a dashboard with heatmaps and line graphs for comprehensive analysis.
9. **Naragain Phumchusri and Phongsatorn Amornvetchayakul (2023)** - Develops customer churn prediction models for a SaaS inventory management company in Thailand facing high churn rates. It compares four machine learning algorithms—logistic regression, support vector machine, decision tree (DT), and random forest—finding that the optimized DT model achieves the best performance with a recall of 94.4% and an F1-score of 88.2%. The study also identifies key features related to churn behavior, providing practical insights for more accurate customer retention strategies and improved decision-making.
10. **Rosario Huerta-Soto and et.al (2024)** -Explores the modernization of the dairy industry, focusing on optimizing supply chain processes with machine learning (ML) to reduce waste and costs. Using PRISMA methodology, it highlights the shift from traditional mathematical modelling to AI/ML-based approaches for better efficiency. The findings show enhanced operational efficiency and improved data transmission during the transit phase. The study also addresses environmental and economic challenges, advocating for advanced optimization methods to increase dairy processing efficiency.
11. **Vikram Pasupuleti and et.al (2024)** - Uses advanced machine learning techniques to enhance logistics and inventory management for a multinational retail corporation. By analyzing historical data, the research improved demand forecasting accuracy by 15%, optimized stock levels, and predicted order fulfillment with 95% accuracy. It also identified at-risk shipments and segmented customers for personalized services. Overall, the application of ML led to a 12% efficiency increase in lead time and an 8% reduction in replenishment errors.
12. **Adedoyin Tolulope Oyewole and et.al (2024)** -Reviews the transformative impact of predictive analytics on supply chain management (SCM), focusing on its applications in demand forecasting, inventory optimization, and supply chain visibility. It explores historical and current uses, real-world case studies, and successful implementations, highlighting the significant improvements in SCM efficiency and decision-making. The study also addresses challenges and limitations in adopting predictive analytics and proposes strategies to overcome them. Looking forward, it discusses emerging trends and technologies, offering insights for practitioners and researchers in this evolving field.
13. **Geetha Manoharan and et.al (2024)** - Presents a novel strategy for optimizing inventory management in e-commerce using predictive analytics and machine learning algorithms. By leveraging historical sales data and relevant factors, the study employs techniques such as decision trees, support vector machines, and neural networks to forecast future product demand. Integrating these predictive models helps e-commerce businesses minimize overstocking and stockouts, enhance operational efficiency, and improve customer satisfaction. The research contributes to the literature on data-driven inventory management solutions and highlights the potential of predictive analytics in addressing e-commerce challenges.
14. **Villacis, Marcelo Y and et.al (2024)** - Explores optimizing sustainable inventory management by integrating advanced data analytics, including Game Theory and GRU models. It highlights the importance of data-driven decision-making in enhancing supply chain sustainability, resilience, and economic viability.
15. **Chawla, V., Itika, I. and et.al (2024)** – To introduce an enhanced ABC analysis method by applying the Pythagorean Fuzzy TODIM approach, addressing the limitations of traditional ABC analysis in handling the imprecision of real-world inventory data. By incorporating fuzzy numbers, the model optimizes inventory levels, reduces costs, and improves operational efficiency. Sensitivity and comparative analyses highlight the model's superiority over conventional methods, demonstrating its reliability and effectiveness in managing uncertain inventory data. This study offers a valuable and innovative approach to inventory management for both practitioners and researchers.

RESEARCH METHODOLOGY

1. DATA INGESTION

Data ingestion is a crucial component of the research framework, laying the groundwork for effective inventory management and forecasting. It encompasses the collection of data from both internal sources—such as historical sales, inventory levels, and demand patterns—and external sources like market trends and economic indicators. After gathering the data, it must be integrated into a unified format to ensure consistency and completeness. This involves merging datasets, aligning formats, and resolving discrepancies. Data cleaning follows, where errors are corrected, missing values addressed, and duplicates removed to enhance data quality. Once cleaned, the data is transformed into a suitable format by normalizing, encoding categorical

variables, and creating derived features. Finally, feature extraction identifies and utilizes relevant attributes like sales volume and demand variability, ensuring that the data used in machine learning models is accurate and meaningful for forecasting and classification.

2. ABC-XYZ CLASSIFICATION ENGINE

The ABC-XYZ Classification Engine is a crucial tool in inventory management, designed to categorize inventory items based on their value and demand variability. The ABC Classification segment ranks items from A to C based on their revenue contribution, with A items representing high-value products that generate a significant portion of revenue, B items having a moderate impact, and C items contributing minimally to overall revenue. Concurrently, the XYZ Classification assesses demand variability using the coefficient of variation (CV), classifying items into X (stable demand with low variability), Y (moderate variability), and Z (highly variable demand). The Integration Module merges these classifications into an ABC-XYZ matrix, categorizing items into combinations like AX, BY, and CZ. This comprehensive approach helps prioritize inventory management efforts by focusing on high-impact items, adapting strategies to demand fluctuations, and optimizing overall stock control.

3. MACHINE LEARNING MODEL

The Machine Learning Model process encompasses several critical steps to ensure accurate inventory predictions and effective management. Initially, data preparation involves splitting the dataset into training and testing sets, normalizing the data, and handling missing values to make it suitable for model training. The model is then trained using machine learning algorithms, such as Random Forest, to learn patterns from historical sales and inventory data. Hyperparameter tuning is performed to optimize the model's performance by adjusting its settings. The model is evaluated with metrics like Mean Absolute Error (MAE) and Mean Squared Error (MSE) to assess its accuracy and reliability. Validation techniques, including cross-validation, are employed to ensure that the model generalizes well to new, unseen data. Once validated, the model is used to make predictions on future inventory needs, providing actionable insights for order quantities and timing. Finally, the model is refined based on performance feedback and new data to enhance its accuracy and adaptability, ensuring that it remains effective for inventory management decisions.

4. FORECASTING AND PREDICTION

The forecasting and prediction process utilizes the trained machine learning model to generate inventory forecasts for future periods. Predictions are made for both six-month and one-year horizons, enabling businesses to plan and manage inventory levels effectively. This forward-looking approach helps in anticipating demand fluctuations and optimizing stock levels, thus reducing the risk of stockouts and overstocking. The forecasts provide actionable insights into future inventory needs, supporting strategic decision-making and enhancing overall inventory management.

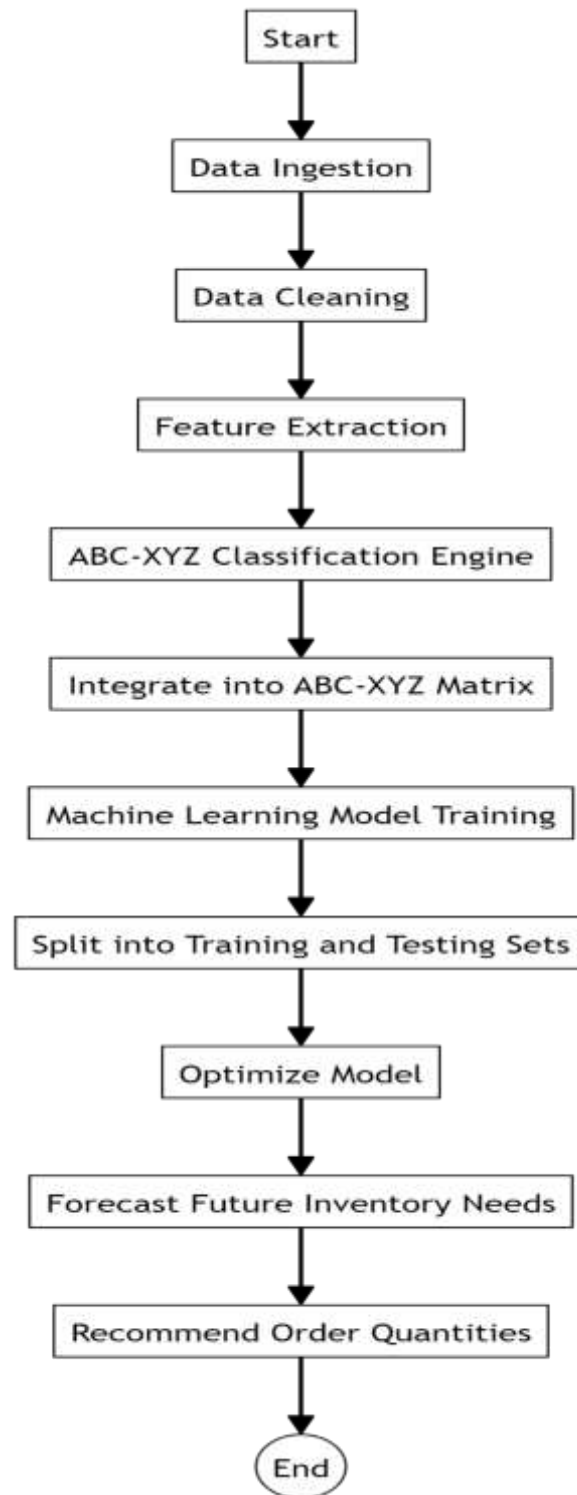


Figure 1: Process Overview

ANALYSIS

Table 1: The ABC Classification of inventory items

ITEM TYPE	TOTAL REVENUE	ABC CLASSIFICATION
Baby Food	\$10,350,327.60	B
Beverages	\$2,690,794.60	C

Cereal	\$5,322,898.90	B
Clothes	\$7,787,292.80	B
Cosmetics	\$36,601,509.60	A
Fruits	\$466,481.34	C
Household	\$29,889,712.29	B

- **A:** High-value items contributing significantly to revenue. Here, Cosmetics and Office Supplies fall into this category, indicating they are key revenue generators and should be managed closely to ensure stock availability.
- **B:** Moderately valuable items that still contribute a substantial portion of revenue but less than A items. Items like Baby Food, Cereal, and Clothes are in this category, suggesting they are important but not as critical as A items.
- **C:** Low-value items with minimal revenue contribution. This includes Beverages, Fruits, and Vegetables, indicating these items are less crucial for revenue but still need to be managed to avoid stockouts.

Table 2: The XYZ Classification of inventory items

ITEM TYPE	XYZ CLASSIFICATION
Baby Food	Z
Beverages	Z
Cereal	Z
Clothes	Z
Cosmetics	Z
Fruits	Z
Household	Z

- **Z:** High variability items with unpredictable demand. All items are classified as Z, reflecting that demand for these items fluctuates significantly. This high variability means inventory planning for these items needs to be more flexible and responsive.

Figure 2: Combined ABC-XYZ Matrix

XYZ Class				
ABC Class		X	Y	Z
	A			B
	B	B	B	B
	C	C	C	C

Figure 2 illustrates the combined ABC-XYZ matrix, integrating both ABC and XYZ classifications. The matrix categorizes items into combinations of ABC and XYZ classifications:

- **AZ:** High-value items with high demand variability. **Cosmetics** and **Office Supplies** are classified here, indicating these are crucial items with unpredictable demand. Management strategies for these items should focus on balancing stock levels to prevent both overstock and stockouts.
- **BZ:** Moderately valuable items with high demand variability. Items like **Baby Food**, **Cereal**, and **Clothes** fall into this category. These items should be monitored regularly and adjusted for demand fluctuations to maintain optimal inventory levels.
- **CZ:** Low-value items with high demand variability. Items such as **Beverages**, **Fruits**, and **Vegetables** are in this category. Although they contribute less to revenue, their unpredictable demand requires careful inventory management to avoid excess stock or shortages.

Overall, the matrix helps prioritize inventory management efforts by identifying which items need more attention based on their value and demand variability. High-value items with high variability require strategic management to optimize inventory levels and ensure availability, while low-value items with high variability need more operational flexibility to adjust to demand changes.

CATEGORY DISTRIBUTION

Table 3: The ABC-XYZ Classification with frequency

ABC-XYZ CLASSIFICATION	FREQUENCY
CZ	6
BZ	4
AZ	2

Figure 3: Frequency of ABC-XYZ Classification



CZ Classification: This is the most common, indicating that many items have high demand variability (Z) and moderate revenue contribution (C). Focused strategies should aim at stabilizing demand for these items.

BZ Classification: Less frequent than CZ but still notable. Items here have moderate demand variability and revenue contribution, requiring balanced management between demand fluctuations and revenue optimization.

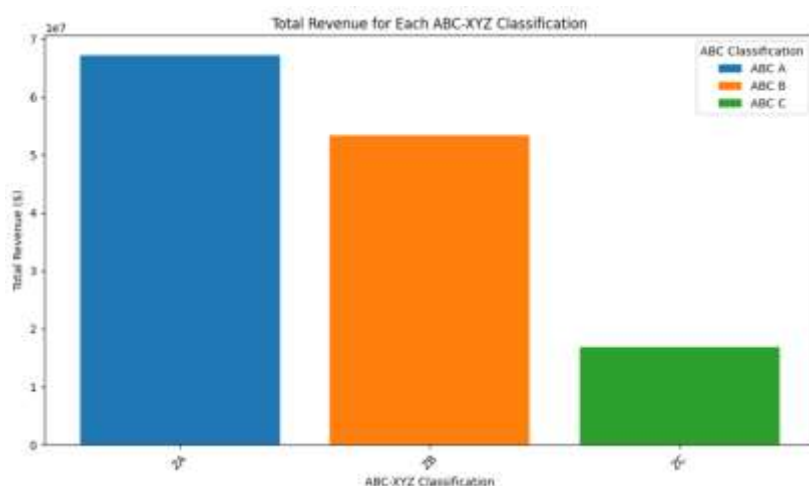
AZ Classification: The least common, representing items with stable demand (A) and high revenue (Z). Although fewer, these items are crucial due to their stable demand and high revenue, necessitating targeted management to maximize profitability.

REVENUE CONTRIBUTION

Table 4: The ABC-XYZ Classification with total revenue

ABC CLASSIFICATION	XYZ CLASSIFICATION	TOTAL REVENUE
A	Z	67186889.67
B	Z	53350231.59
C	Z	16811647.05

Figure 4: Total Revenue of ABC-XYZ Classification



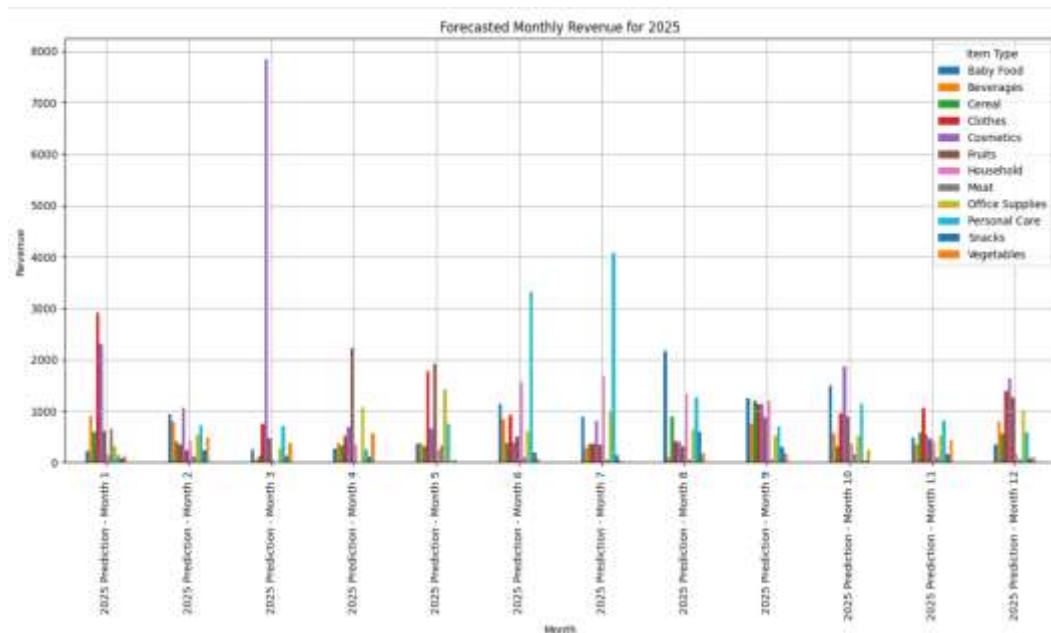
AZ Classification: This combination, representing items with high revenue contribution (A) and stable demand (Z), has the highest total revenue. Items in this category are crucial for inventory prioritization due to their significant impact on revenue.

CZ Classification: Despite its moderate revenue, CZ classification is common among the items. Effective inventory management should focus on stabilizing demand for these items to improve overall performance.

BZ Classification: This category, with moderate revenue and demand variability, shows balanced revenue contribution. It requires a strategic approach to manage both demand fluctuations and revenue optimization.

FORECASTING FUTURE SALES

Figure 5: Forecasted Revenue for 2025



Using a Random Forest Regressor, we forecasted sales for each item type over the next 12 months based on historical data. The forecast reveals significant revenue patterns and trends, showing items like Cosmetics and Office Supplies with consistently high predictions, indicating stable demand. In contrast, categories such as Baby Food display fluctuating sales, suggesting varying demand throughout the year. This forecast helps in strategic inventory management by identifying high-revenue periods and allowing for adjustments in stocking and promotional efforts to align with anticipated demand.

To optimize inventory in retail and e-commerce settings based on the analysis, start by leveraging insights from category distribution, forecasted sales, revenue distribution, and ABC-XYZ classification. Focus on high-revenue categories and ensure that inventory levels are aligned with forecasted sales. Increase stock for items with high forecasted demand and reduce it for those with lower demand. Maintain safety stock for items with variable demand to avoid stockouts. Utilize ABC classification to prioritize inventory management: ensure high-value (Class A) items are well-stocked and monitored closely, manage moderate-value (Class B) items efficiently, and reduce stock for low-value (Class C) items. Apply XYZ classification to handle demand variability: keep stable-demand items (Class X) well-stocked with predictable ordering, adjust stock levels flexibly for items with moderate variability (Class Y), and maintain higher safety stock for items with high variability (Class Z). Regularly review and adjust inventory levels based on changing sales patterns and forecasts. Implement advanced inventory management systems and tools tailored for retail and e-commerce environments, such as automated reordering systems and real-time analytics. These strategies will enhance inventory efficiency, reduce costs, and ensure that appropriate stock levels are maintained to meet customer demand effectively in both retail and e-commerce contexts.

CONCLUSION

A comprehensive approach to optimizing inventory management through the integration of ABC-XYZ classification and machine learning techniques. By leveraging historical sales data, inventory levels, and demand patterns, we developed a robust system that classifies inventory items based on their revenue contribution (ABC Classification) and demand variability (XYZ Classification). The ABC Classification categorizes items into three groups: A (high-value items), B (moderate-value items), and C (low-value items), enabling businesses to focus resources on high-impact products. The XYZ Classification evaluates demand stability, with categories X (stable demand), Y (moderate variability), and Z (highly variable demand). The combined ABC-XYZ matrix provides a dual perspective on item importance and demand predictability, guiding more effective inventory management strategies. Our machine learning model, employing algorithms such as Random Forest, was trained on historical data to forecast future inventory needs. The model's accuracy was validated through performance metrics, ensuring reliable predictions for inventory management. This forecasting capability is crucial for maintaining optimal stock levels, reducing excess inventory, and preventing stockouts.

The analysis of category distribution, forecasted sales, revenue distribution, and the detailed ABC-XYZ classification enhances our understanding of inventory dynamics. These insights enable businesses to prioritize high-impact items, adjust strategies based on demand variability, and optimize stock levels. Future enhancements include the development of a web application to expand the model's accessibility and facilitate Power BI report generation for advanced analytics. Additionally, analysing seasonal trends for individual items will further refine predictions and align inventory strategies with varying demand patterns throughout the year. Overall, this research contributes valuable methodologies for inventory optimization in retail and e-commerce sectors, offering actionable insights to improve stock control and operational efficiency.

FUTURE ENHANCEMENTS

Web Application: Create a user-friendly web application that integrates the inventory optimization model. This app should allow users to input data, run the model, and generate Power BI reports for detailed analytics. The web app should include features such as interactive dashboards, real-time data updates, and customizable reports. It will provide a scalable solution for users across different organizations to leverage the model's insights, making it accessible and easy to use.

Seasonal Trends: Extend the model to incorporate seasonal analysis for individual items. While you have analysed normal historical data, adding a seasonal component will enhance the accuracy of forecasts and inventory management. This involves examining how seasonal factors, such as holidays or weather changes, affect the demand for different items. By integrating this analysis, you can refine inventory strategies to account for peak seasons and fluctuations, improving stock levels and reducing overstock or stockouts.

REFERENCES

1. DHAYALAN, V., et al. "SAVING HABIT AND INVESTMENT PREFERENCE OF GOVERNMENT SCHOOL TEACHERS IN VELLORE DISTRICT."
2. Dr.P.Venkatesh (2021). *A Study on School Children's Perception Towards Frequency Of Visits And Selection Of Junk Food Retail Chain Store*. *Natural Volatiles & Essential Oils*, ISSN: 2148-9637, Vol.8 (5), 8181-8190.
3. Dr.P.Venkatesh (2021). *A Study On Technical Analysis Of The Commodity Market With Special Reference To Gold And Silver*. *Natural Volatiles & Essential Oils*, ISSN: 2148-9637, Vol.8 (4), 366-379.
4. Dr.P.Venkatesh (2021). *A Study on Technical Analysis Using Candlestick Pattern of Selected Large Cap Stocks Listed in National Stock Exchange (NSE), India with Reference to Steel Sector*. *GSI Journals Serie B: Advancements in Business and Economics*, 3 (2), 62-71. DOI: 10.5281/zenodo.4972495
5. Dr.P.Venkatesh (2021). *A Study on the Weak Form Efficiency of Metals & Mining Sector in BSE*. *Natural Volatiles & Essential Oils*, ISSN: 2148-9637, Vol.8 (4), 287-296.
6. Dr.P.Venkatesh (2021). *Study on Price Discovery of Selected Indian Agriculture Commodity with Special Reference to NCDEX*. *International Journal of Advance and Innovative Research*, ISSN: 2394-7780, Vol.8 (3), 76-84.
7. K. Maran, C. R. Senthilnathan, S. Usha and P. Venkatesh, "Business Analytics Contribution in the Growth of Indian Digital Business," 2022 1st International Conference on Computational Science and Technology (ICCST), CHENNAI, India, 2022, pp. 497-500, doi: 10.1109/ICCST55948.2022.10040343.
8. Krishnamoorthi, M., Guha, D., Nikitha Neelappa, S., Shaikh, A., Venkatesh, P. (2025). Testing the Weak-Form Market Efficiency of IT Sector: An Empirical Evidence from India. In: Hamdan, A., Braendle, U. (eds) *Harnessing AI, Machine Learning, and IoT for Intelligent Business*. *Studies in Systems, Decision and Control*, vol 555. Springer, Cham. https://doi.org/10.1007/978-3-031-67890-5_56
9. M. K, S. C.R., U. S and V. P, "Impact of Solar Energy on Mitigating Climate changes for a Sustainable Development in India," 2022 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2022, pp. 1-5, doi: 10.1109/ICPECTS56089.2022.10046744.
10. Maran, K., et al. "Business Analytics Contribution in the Growth of Indian Digital Business." 2022 1st International Conference on Computational Science and Technology (ICCST). IEEE, 2022.
11. Maran, K., et al. "Impact of Solar Energy on Mitigating Climate changes for a Sustainable Development in India." 2022 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS). IEEE, 2022.
12. Maran, K., Prabakaran, T., Priyadarshini, P., & Venkatesh, P. (2022). IMPACT OF ENUMERATION FACTORS ON EFFECTIVE SELF HELP GROUP (SHG) FUNCTIONING. *International Journal of Early Childhood Special Education*, 14(3).
13. Maran, K., Priyadarshini, P., Jenifa, L., Senthilnathan, C. R., & Venkatesh, P. (2021, December). *Data Analysis on Mobile Payment Technology with Reference to Users' Behaviour of Retail Goods in India*. In 2021 4th International Conference on Computing and Communications Technologies (ICCCCT) (pp. 267-272). IEEE.
14. Murugan, K., and P. Venkatesh. "Viral Marketing of Digital Products Using Social Media." *International Journal of Information Technology & Computer Sciences Perspectives* 2.1 (2013): 120.
15. Murugan, K., S. Helen Roselin Gracy, and P. Venkatesh. "A Study on Customer Perception Towards Mugi Ultra Liquid Detergent." *IJMPERD* 8.3 (2018): 978-984.

16. Murugan, K., Selvakumar, V., Venkatesh, P., Manikandan, M., Ramu, M., & Krishnamoorthi, M. (2023, December). *The Big Data Analytics and its Effectiveness on Bank Financial Risk Management*. In *2023 6th International Conference on Recent Trends in Advance Computing (ICRTAC)* (pp. 313-316). IEEE.
17. Venkatesh, Dr P. "A Study on School Children's Perception Towards Frequency Of Visits And Selection Of Junk Food Retail Chain Store." *Natural Volatiles & Essential Oils*, ISSN (2021): 2148-9637.
18. Venkatesh, P. S. K. P. S., Krishna SUDHEER, and Senthilmurugan Paramasivan. "A study on technical analysis using candlestick pattern of selected large cap stocks listed in National Stock Exchange (NSE), India with reference to steel sector." *GSI Journals Serie B: Advancements in Business and Economics* 3.2 (2021): 62-71.
19. Venkatesh, P., and D. S. Revathi. "A Study on Performance Analysis of Selected Mutual Fund Schemes in India." *Solid State Technology* 63.2S (2020).
20. Venkatesh, P., et al. "A Comprehensive Investigation to Examine the Preferences and Satisfaction Levels of Outpatients in Relation to the Quality of Services Provided by Hospitals in the Vellore District." *2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS)*. IEEE, 2023.
21. Illakya, T., Keerthana, B., Murugan, K., Venkatesh, P., Manikandan, M., & Maran, K. (2024). *The role of the internet of things in the telecom sector*. *2022 International Conference on Communication, Computing and Internet of Things (IC3IoT)*, 21, 1–5. <https://doi.org/10.1109/ic3iot60841.2024.10550390>
22. Manikandan, M., Venkatesh, P., Illakya, T., Krishnamoorthi, M., Senthilnathan, C., & Maran, K. (2024). *The Significance of Big Data Analytics in the Global Healthcare Market*. *2022 International Conference on Communication, Computing and Internet of Things (IC3IoT)*. <https://doi.org/10.1109/ic3iot60841.2024.10550417>
23. Ilakkiya, T., Manikandan, M., Ch, R. K., M, K., Ramu, M., & Venkatesh, P. (2024). *Neuro Computing-Based Models of Digital Marketing as a Business Strategy for Bangalore's Startup Founders*. *Ieee*, 1–3. <https://doi.org/10.1109/incos59338.2024.10527779>
24. Venkatesh, P., Selvakumar, V., Ramu, M., Manikandan, M., & Senthilnathan, C. R. (2023). *Measure of Well-Being of Freelancers in it Sector*. *Ieee*. <https://doi.org/10.1109/iccebs58601.2023.10448738>
25. VENKATESH, P., et al. "A STUDY ON IMPACT OF PROFIT, EARNING PER SHARE AND DIVIDEND ON EQUITY PERFORMANCE ON SELECT STEEL SECTOR USING DISCRIMINANT FUNCTION ANALYSIS."